MIAI Cluster IA

Dépôt de projets dans le cadre de l'appel à chaires 2025-2026

Application form / Formulaire de soumission

I. Project identification, Summary & Key-words

Title of the project : Federated learning and organization in mobile agent societies

Acronym of the project: FLOMAS

Summary (10 lines): We consider a group of autonomous, mobile cyber-physical agents that embed both communication and computation capabilities. These agents operate according to a set of organisational rules that coordinate their sensing and actuation, allowing a social structure to emerge spontaneously. Each agent runs a lightweight learning algorithm and exchanges only the essential parameters needed for collaboration. Distributed protocols continuously adjust organisational roles and communication strategies in real time, preserving overall efficiency despite the agents' mobility. By adopting a clustered federated learning method, the system exploits locally collected data to personalise the artificial neural networks to its environment. Because the agents are linked through temporal graphs, they can self-organise and adapt online, opening new avenues for distributed machine learning methods that provide more tailored and efficient learning within ever-changing groups.

5 Key-words: distributed learning, self-organization, social organizations, multi-agent systems, cyberphysical systems

II. Context and scientific description (6 pages)

- *Context, objectives and importance of the problem addressed (1 page)*

Cyber Physical Systems (CPS) become ubiquitous: smart cities, smart grid, swarms of drones, automated factories, cooperating autonomous vehicles, etc. Their large-scale deployment relies on fine and reactive coordination of multiple subsystems, which are often distributed, heterogeneous, and dynamic. Traditional centralized approaches — where a single supervisor makes all decisions — reach their limits when facing scalability, resilience, latency or confidentiality issues. Multi-Agent Systems (MAS) offer an ideal theoretical and practical framework to address these challenges [Jamont 2015, Januário 2019, Kouicem 2020]. By distributing cooperation among autonomous agents with loosely coupled decision loops, the MAS approach allows to coordinate agents with limited local knowledge to achieve coherent global objectives.

To tackle this global coordination problem, we are interested in giving self-organizing capabilities to groups of CPS by providing them with learning abilities directed towards the dynamic structuration inside and between the groups. This dynamic structuration will be studied using the concept of adaptive organization [Mintzberg 1979, Ferber 2004, Sun 2025]. In particular the aim is to use the property of emerging adaptive organizations to guide task delegation, role assignment, and conflict resolution among the CPS, as well as their ability to identify each other and cooperate at the global level.

In this context, our study will consider an open network of autonomous mobile agents – such as sensors/actuators, robots, aerial drones, or unmanned surface vessels – connected through a wireless ad hoc network. The communication topology is time-varying since agents may decide to connect/disconnect with their neighbours as they move in the environment (possibly outside the communication range). These agents have limited sensing/actuation, communication and computing embedded capabilities. However, their cooperation following specific organizational rules may allow them to achieve complex tasks such as trajectory/path planning, rendez-vous, deployment and operation of a local communication network, regional sensing/actuation tasks, etc.

The project central research will focus on the design of adaptive self-organization and federated (distributed) learning algorithms. Two issues will be addressed: learning social organization for CPS andtraining a CPS to perform a task through the evolution of its social organization. These issues will need to consider the following scientific challenges:

- SC1 : Identify organisational models (using top-down, bottom-up or mixed approaches, [Liu 2025]) and agent capabilities suitable for federated learning [McMahan 2017] and emerging adaptive social organization, including human defined constraints/expertise. This includes the dynamic structuration of the groups in a decentralized manner.
- SC2 : Build ad-hoc communication temporal networks with distributed routing algorithms (see e.g. [Hamani 2018, Nouali 2025]) where mobile agents may decide locally to connect/disconnect with neighbours and groups may decide to merge/split temporarily to achieve collective tasks. The challenge will be to guarantee global properties on the communication network, such as connectivity, while local modifications occur [Atman 2021, Venkateswaran 2024]. The developed decentralized algorithms will be based partly on the information theoretic local analysis of the communication time series [Toupance, 2024] and regional reachability/observability analysis [Yacoubi 2021]
- SC3 : Detect communities of agents that have learned similar behaviours in order to personalize their artificial neural network (ANN) to their environments is a hot challenge in clustered federated learning (CFL) [Ghosh 2020]. Data confidentiality often prevent the data to be shared. Yet, the most efficient ANN comparison methods rely on comparing ANNs output or ANNs activation [Kornblith 2019]. A first challenge is therefore to find an efficient approach to compare policies without any data.
- SC4 : Personalize ANNs in a way that they are the right compromise between an over-generalization of the task or an over-specialization on the local data [Bettinelli 2025]. These challenges are paramount in standard federated learning, but they are even more complicated in reinforcement learning with where agents might learn their policy using several ANNs.

- *Impact, structuring effect on our ecosystem as well as environmental/societal impact (1/2 page)*

The FLOMAS Chair will provide an opportunity to consolidate the collaboration between LISTIC (USMB) and LCIS (UGA) researchers and federate their respective academic and industrial partners around the theme of multi-agent approaches and federated learning applied to cyber-physical systems. The local academic partners are within the LIMOS (axes *Decentralized and Open Socio-Cognitive Systems* and *Multiagent simulation*, both Saint-Etienne), LIRIS (axes *Multiagent systems SysCosMA* et *Data Mining and MAchine Learning*, Lyon) and LIRMM (*Marine Robotics* team, Montpellier) laboratories.

FLOMAS will be complementary with the ESAIM-CPS Chair, currently held by I. Prodan and L. Morge-Rollet at LCIS: FLOMAS will focus on higher level decision making algorithms, aimed at determining agents' current goals and coordination strategies while ESAIM-CPS is focused at lower-level, constraints aware, safe and secure control strategies. A thematic link can also be found with the Edge Intelligence Chair (https://edge-intelligence.imag.fr/), led by Denis Trystram and initially focused on distributed learning (it has now shifted toward low-complexity, distributed, and frugal methods), our contribution being more strongly shaped by the constraints inherent to embedded systems, which will condition and influence every stage of the agents' development lifecycle.

Among the applications targeted from the outset of the project are the monitoring of natural aquatic environments (including pollution detection and tracking), searching for and rescuing victims in remote/dangerous areas, as well as forest fire spread detection and monitoring. Furthermore, the proposed approach ofr autonomous deployment of ad-hoc communication networks in CPS meets requirements for reliability/resilience, trust and data security, autonomy and delegation of responsibility. Finally, the proposed design method for adaptive social organisations is also in line with the desire to develop frugal solutions that minimise energy consumption, the volume of data exchanged and the centralisation of calculations (edge computing). Therefore, we believe the FLOMAS chair research proposal addresses clearly identified environmental and societal challenges

- *Scientific challenges, research directions, work packages and tasks (3 pages)*

To address scientific challenges SC1 to SC4, we have identified the following research directions:

RD1: Identify and characterize organizational elements (OEs) - such as roles, groups, interaction protocols, norms, objectives, *etc.* - which are relevant both for the tasks and CPSs constraints.

RD2: Analyse in a population of trained CPS, the variety of produced individual and collective behaviours to identify regularities that could correspond to OEs.

RD3: Focus the learning process towards policies constrained by role-related OEs, and interactions to produce collective behaviours.

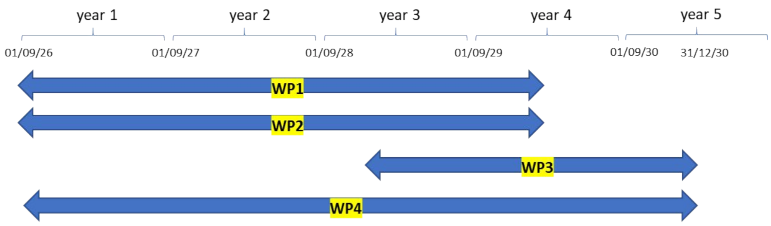
RD4: Explore clustered federated learning aggregation methods to personalize agents policies to their environment.

RD5: Embed complementary constraints on the role-related OE assigned to agents in order to prepare the possibility for a human operator to impose some roles on selected agents.

The FLOMAS chair is structured as follows. As part of WP1, distributed algorithms are being developed to enable a collective of autonomous agents to self-organise in order to perform various tasks in a changing environment. As part of WP2, CFL methods are being developed to enable groups of CPS to autonomously cover harsh terrain while maintaining communication between agents. WP3 is mostly dedicated to integration and application to simulation benchmarks and life-size search missions in the field, with the Haute-Savoie fire and rescue service (SDIS 74). WP4 is dedicated to project coordination and management, as well as dissemination of the results.

The chair research team consists of five permanent researchers, three from LCIS and two from LISTIC: L. Lefèvre (chair holder, LL, LCIS), M. Bettinelli (chair co-holder, MB, LISTIC), C. Raiëvsky (CR, LCIS), L. Duvicier (LD, LCIS) and V. Couturier (VC, LISTIC). Two students PhD1 (Distributed Adaptive Organization, LCIS) and PhD2 (Clustered Federated Learning, LCIS) and one postdoctoral researcher (PD, LCIS/LISTIC) will contribute to the research project, as well as permanent staff outside the FLOMAS chair team: A. Desuert (AD, research engineer, LCIS), J.-P. Jamont (JPJ, professor LCIS), P.-A. Toupance (PAT, senior lecturer, LCIS) and A. Benoït (AB, professor, LISTIC).

Person-month load					
	WP1	WP2	WP3	WP4	Total
LL (CH)	9	6	3	2	20
MB (CoH)	6	9	3	1	19
CR	6	3		1	10
LD	3	3	3	1	10
VC		3	6	1	10
PhD1	26	6	3	1	36
PhD2	6	26	3	1	36
PD	4	4	15	1	24
Total	60	60	36	9	
Total person-months for the project					165



Contributions of the chair team members (PM) and time distribution of the work-packages

Work Package 1: Social organization emergence and adaptive dynamics

WP Leader: LCIS. **Participants:** MB, LD, LL, CR, PhD1 . **External participants:** JPJ, PAT

Objectives: developing decentralised algorithms allowing groups of CPSs to dynamically self-organize in order to stay efficient in changing conditions and missions.

Description: Open groups of CPSs face complex and dynamic environments which require the ability to modify their behaviours, roles, and interactions in order to continue to carry on their tasks. Maintaining an organization that supports the group's mission is a demanding challenge. Designers of MAS in CPSs contexts inevitably face the overwhelming complexity of the state-action space. To overcome this limitation, WP1 will introduce learning capabilities at the MAS's organizational level.

The study of RD1 will consist in identifying OEs relevant to i) the class of missions targeted by the chair and, ii) the constraints of the targeted CPs, using expert knowledge. In RD2, the OEs will be identified by an analysis of the behaviours produced by a "raw" MARL process applied to the task. The variety of produced policies will be analysed to identify regularities that correspond to OEs. These regularities will be identified either by analysing the trajectories of the policies in the state-action space, as proposed in [Soulé2025a, Soulé2025b], or by utilizing tools from Federated Learning [Bettinelli 2025] to identify and merge sufficiently similar policies. This last type of analysis will also be used in WP2 and will therefore be an interaction point between the two first WP.

The comparison of the behaviours produced by these two research directions, RD1 and RD2, will give us insight in the relative advantages of the produced policies and used processes. In RD3, the OEs identified by the first two RDs will be utilized to focus the learning process towards policies related to the group's organization in a refinement phase. This phase will aim at producing individual behaviours, constrained by role-related OEs, and interactions to produce organization-oriented collective behaviours.

Once these collective behaviour obtained, RD5 will be explored by extending the training in setups where arbitrary, forced, constraints on the role-related OEs are assigned to agents in order to prepare the possibility for a human operator to impose some roles on selected agents. In this situation, the behaviours of the other agents have to adapt to these imposed constraints.

The main deliverable of WP1 will be methodology and a set of tools allowing the designer of a CPS to identify relevant OEs from an automated analysis of “raw” policies and to refine these policies using the identified OEs as constraints to obtain organization-oriented policies.

Work Package 2: Adaptive Collaborative Control of CPS through Clustered Federated Reinforcement Learning

WP Leader: LISTIC. **Participants**: MB, VC, LD, LL, CR, PhD2**. External participant:** AB

Objectives: Conceive a framework that make agents learn and share their knowledge in harsh environments.

Description: WP2 focuses on developing a scalable, energy-efficient learning framework that enables groups of CPSs to autonomously cover irregular terrains while preserving inter-agent communication. We assume that groups might learn and be used in very different environments (mountains, lakes and other terrains) with a wide variety of obstacles. However, standard FL aggregates all local updates into one global ANN. When agents have specific characteristics, this aggregation might be over-generalized and may not satisfy any agent’s operational constraints. Recent advances in CFL [Zhang 2021] address this point by automatically detecting groups of participants with similar data distributions and training a dedicated ANN for each cluster. This allows groups to generalize their knowledge to their operating environment without learning from environments that are too dissimilar, as learning from such different environments would be harmful to the learning process. However, CFL combination with RL (CFRL) [Fan 2021] is still a very young research field and more research must be conducted for the specific problem of cooperative area-coverage in harsh environments. WP2 focuses on three research directions: **RD2**, **RD4** and **RD5**.

RD2 addresses the question of policies similarity. Comparing policies requires to compare artificial neural networks (ANN), which often requires data to assess the difference of output or neuron activation of two ANNs. However, standard FL framework does not allow agents to share their data. CFL approaches therefore compare ANN parameters to assess if two ANN produce the same behaviour. One of the most common method is to process the Euclidean distance of two ANN. However, this method is subject to the curse of dimensionality which greatly reduce is efficiency. Thus, there is a need to test and identify methods that are the most adapted to compare ANNs. This work will benefit both WP1 and WP2.

RD4 tackles ANN personalization problem. The most intuitive approach for aggregating similar ANNs is to average them parameter by parameter [McMahan 2017]. But the side effect is to force the generalization of agents policies, without letting them the chance to specialize their own ANN to their actual environment. Agents could take advantage of their autonomy to decide by themselves is they need to learn from others or not.

RD5 addresses the challenge of adapting CPSs to human intervention. Building on the collaborative learning capabilities developed in WP2, CPSs not directly operated by a human could learn to adjust their policies in response to unexpected events, such as a human action on one or several CPSs.

The main task of WP2 is to conceive the CFRL framework to make groups learn in harsh environments. As for WP1, it involves formalizing the coverage and communication objectives but also designing a method able to take into account client clusters during the FL training, those clusters being known in advance as a first step then discovered automatically. This framework will be the deliverable of this work package.

Work Package 3: Integration and experimental use cases

WP Leader: LISTIC. **Participants:** MB, VC, LD, LL, Post-doc PD. **External participants:** AB, AD

Objectives: Integrate methods developed in WP1 and WP2 in real-life use cases.

Description: : WP3 focuses on integrating and evaluating the theoretical contributions of WP1 and WP2 within realistic operational scenarios. Building on the organisational principles established in WP1 and the Clustered Federated Reinforcement-Learning adaptation mechanism developed in WP2, WP3 aims to combine these approaches into a unified learning framework capable of tailoring agent behaviours to the specific characteristics and constraints of diverse environments. A post-doctoral researcher will be responsible for implementing this integration and conducting systematic evaluations on the use cases defined with our partners.

Our consortium benefits from strong collaborations with SDIS 74 (Haute-Savoire fire and rescue service), ODAS Solutions (embedded electronics) and CT2MC (Unmanned Surface Vehicles), each providing real-world operational contexts that will stress-test the proposed methods. SDIS 74 provides a scenario in which a fleet of drones is deployed for search-and-rescue missions in mountainous terrain. Currently, these drones are not autonomous and rely on offline data processing, making this scenario an ideal initial application for the proposed adaptive capabilities. ODAS Solutions, which operates aerial drones and drone-boats (Bathydrones), contributes both technical expertise and extensive field experience. Together with LISTIC, the company has developed the OPEN RESCUE solution -already deployed by SDIS 74- which uses ANNs trained on more than 150,000 labelled images to support missing-person search missions. Enhancing this system with inter-drone collaboration is a key research objective of the project, particularly for navigating complex environments and covering large search areas. ODAS also provides datasets enriched with terrain information and drone geolocation data, enabling early simulation of multi-agent collaboration and federated learning before real-world testing.

CT2MC, specialised in unmanned surface vehicles, and offers a use case involving lake-pollution monitoring with a fleet of USVs communicating through an ad-hoc short range RF communication network, one of the boat being cable-connected with an underwater robot. The supervision of this fleet being the main subject the research project ROSEAU (project ANR PRCE from the French National Research Agency, which includes funding for a two-year research engineer), WP3 will focus mostly on the search and rescue use case with SDIS 74 and ODAS, only supervising the implementation of the algorithms developed in the chair to the CT2MC USVs use case.

The WP3 evaluation plan consists of three stages. First, the integrated framework will be assessed in simulation for each use case using ROS 2 and Gazebo, measuring key metrics such as environment coverage ratio, completion time, adaptation time after terrain changes, and communication connectivity within the fleet. Second, the same metrics will be validated on physical CrazyFlie drones available at the LCIS laboratory to ensure compatibility with embedded hardware. Finally, the fully integrated system will be deployed on the SDIS 74 fleet for a field trial, testing the entire approach under authentic rescue conditions and demonstrating its operational viability.

Work Package 4: Management and dissemination

WP Leader: LCIS. **Participants:** MB, VC, LD, LL, CR, PhD1, PhD2, PD

Objectives: Handle administrative and dissemination objectives

Description: WP4 is subdivided into the following tasks:

– **general project management:** organizing the exchange of information (website/share folder), regular project general/technical meetings, the mobility of PhD an PD students, etc.
– **project results dissemination** in academic and industrial venues: organizing the project regular workshops (3), invited sessions in conferences (1-2) and special issue (1), demonstration organised with partners and their customers / external partners, outreach activities for the general public (a.o. the annual science festival)
– **technical coordination:** ensuring software interoperability and hardware compatibility, availability of the deliverables , synthesising the project outcomes, and anticipating the following projects

- *Team: synergies and collaborations (1/2 page)*

The FLOMAS team is multidisciplinary, combining a balanced mix of two young (MB/CoH, LD), two intermediate (CR, VC) and one senior (LL/CH) researchers with expertises that align perfectly with the project's goals: complex systems modelling (LD), analysis (LL) and distributed control (LL), multiagent systems (CR, MB), organizations (LL, CR), agent-based information systems (VC), and federated learning (MB, VC). Chair holder and coholder have also experience with core applications related to mobile sensor networks, UAV swarms and USV fleets, among other through joint projects with industrial and civil society partners (SDIS 74, CT2MC, ODAS) This diverse skill set enables the team to address all aspects of the project, from novel research and technological innovation to educational impact. The team works within a rich ecosystem of research at the LCIS and LISTIC labs and educational activities at IUT Valence (BSc), IUT Annecy (BSc) and Grenoble INP – Esisar (MSc). Besides the chair team, additional external academic members will provide their work and expertise to the project, including Prof. J.-P. Jamont (LCIS, embedded multiagent systems); Prof. A. Benoit (LISTIC, semantic analysis of data, federated learning) and Dr. A. Desuert (LCIS research engineer, embedded software, ROS integration). The FLOMAS chair will also benefit from the Esisarium platform at LCIS with its nanodrones, computing facilities and optical localisation system.
The chair will also benefit from the academic network of the members, among others the IXXI national research center on complex systems (Lyon), LIMOS groups on *Open Socio-Cognitive Systems* and *Multiagent simulation (*Saint-Etienne), the LIRIS *Multiagent group* (Lyon) and the LIRMM *Marine Robotics* team in Montpellier. Also Thalès Land & Air Systems France will participate separately in the project (see innovation, part, hereafter), as well as colleagues from the computer department at Unige (temporal graphs, Geneva) and the University of Florence (Center for the Study of Complex Systems).

Finally, the chair consortium plans to actively participate in the creation of a new national Groupement d'Intérêt Scientifique (GIS) entitled "*AI for Civil Security and Extreme Situations*". This GIS aims to bring together academic laboratories, industrial partners, and civil-protection organisations to coordinate research efforts, share datasets and experimental infrastructures, and jointly develop AI-based tools for safety-critical scenarios. By engaging in the early stages of its construction, our project will help shape the national research agenda in this domain, strengthen collaborations beyond the current consortium, and ensure long-term dissemination and uptake of the methods developed in WP1–WP3. This involvement will reinforce the project's scientific impact at the national scale and anchor its contributions within a broader, durable research network.

- *Description of how the budget will be allocated, and description of co-funding*

The project budget totals 444k€: **i)** 390 k€ for salaries, including PhD1 (hosted mostly at LCIS/UGA, recruited in October 2026), PhD2 (hosted mostly at LISTIC/USMB, recruited in October 2026), and one two-years PostDoc (PD) position (LCIS/LISTIC, recruited in December 2028); **ii)** 42k€ for operating costs, covering internships (12k€ for 4x5 months), international missions (12k€ for 6 mobilities), coordination meetings/workshops (12 k€) and 1-months research stays (6k€ for 4 stays) ; **iii)** 12 k€ for equipment including laptops (6k€ for 3 units), PCs with AI-enabled boards (6k€ for 2 units).

Co-funding: LCIS and CT2MC already collaborated (joint PhD supervision) and are currently preparing an ANR-PRCE proposal on the monitoring of aquatic environment with fleets of USV/UUV which includes a two-years PD position (130 k€), a two-years Development Engineer position (150k€) and USV (20 k€). LSTIC, SDIS74 and ODAS already collaborated and the FLOMAS PD position will help them to improve their search and rescue solution, while they will provide the full drone system, data, use cases and organise full-scale validation during their regular simulation exercises. The ESISARIUM plateform at LCIS/Esisar will be used for lab experiments with its fleet of nanodrones, its localization systems and the development environment. The collaboration with Thalès Land & Air Systems France will be the object of a separate collaboration contract (anti-drone measures, enactive physically informed sensor networks) with NDA and IP clauses which includes PhD joint supervision.

- *Sustainability, long-term strategic vision*

The FLOMAS chair addresses important societal issues such as environmental monitoring, victim localisation, anti-drone warfare, ... Besides, it aims at developing scalable, resilient, energy efficient strategies for adaptive coordination and cooperation of autonomous CPSs which are becoming ubiquitous. Therefore the chair scientific directions are most likely to be relevant for the next decade.

The chair's regular meetings will be extended to workshops, open to the chair's researchers' industrial and academic partners, which will provide an opportunity to structure the regional community around multi-agent systems for CPS. These workshops will also enable young doctoral students and interns to meet and present their work.

A graduate school — intended to continue beyond the chair time horizon — will be organised during the chair's funding period to expand this initial core group. The community thus formed will then seek to develop through proposals for guest sessions and special issues at conferences (JFSMA, NECSYS) and in national (ROIA) and international (from IFAC and IEEE SMC/CSS societies) journals. The researchers in the chair will contribute with their initial academic network including colleagues from the region (LIRMM, LIMOS, LIRIS), from France (Cristal lab in Lille, IMT Nord Europe in Douai, ISAE Supaero in Toulouse, UPVD in Perpignan) and Europe (University of Geneva, TU Munich, University of Florence). This network, already strongly connected at the regional level, will allow to prepare proposals for the next French and European calls, jointly with industrial and civil society partners.

III. Innovation (1-2 pages)

Research Chair Partnerships and Objectives

The chair brings together two private partners, ODAS Solutions (embedded electronics) and CT2MC (Unmanned Surface Vehicles), and one public partner, SDIS 74 (Haute-Savoire fire and rescue service). Each of these partners has provided a letter of support outlining their commitment and expected contributions. The society Thalès Land & Air Systems (LAS) will cooperate separately with chair members for confidentiality and intellectual property issues.

For all three core-partners, the research program of the chair is considered essential to the advancement of their innovation strategies.

- For **ODAS Solutions**, a key objective is to enhance inter-drone cooperation and drone intelligent capabilities.
- For **SDIS 74**, the goal is to improve the performance of the **OPEN RESCUE** system for locating missing persons in complex environments.
- For **CT2MC**, the collaboration focuses on the coordination of a fleet of USVs and UUVs for aquatic environment monitoring.

Nature of Partnerships and Associated Funding

The partnerships extend beyond the academic sphere, with potential modalities including internships, CIFRE (work-study PhD) contracts, and joint applied research projects. Having already collaborated with all involved partners, we are confident that this initiative will foster fruitful exchanges both scientifically and in terms of technology transfer. The inter-drone cooperation and improvement of the OPEN-RESCUE system are directly connected to WP3 in the chair proposal. The post-doc research position in the chair (PD) will mostly work on the implementation of the algorithms to this use case, helped with engineers from ODAS and (for real-life tests) by SDIS 74 firemen. The collaboration with CT2MC involves two engineers from the company, as well as a new research engineer (24 months) and a post-doc researcher (24 months) funded by the ANR project ROSEAU. The collaboration with Thalès LAS will be the object of a separate contract that provides for the recruitment of PhD student (CIFRE work-study program).

Expected Outcomes

- **ODAS Solutions**: Founded in Annecy in 2023, ODAS is a small company of seven employees. It manufactures drone-boats and distributes aerial drones from various brands. Its ambition is to deliver innovative solutions to clients, particularly systems based on machine learning and deep learning, and to provide the drones it produces or resells with intelligent capabilities. The chair will enable ODAS to explore fundamental research topics that the company cannot pursue in its daily operations. By doing so, ODAS aims to gain a competitive edge in the development of intelligent drone swarms. Since its creation, the company has sought to maintain strong ties with research, allowing it to position itself on high value-added, cutting-edge service missions. Dissemination of the research results will further strengthen ODAS's scientific and technical reputation.
- **SDIS74**: The emergency services of Haute-Savoie operate one of the largest drone fleets among rescue organizations, with 40 certified pilots. Their objectives are to:
  - enhance the performance of the **OPEN RESCUE** system by increasing both the detection rate and the detection speed for missing persons, while ensuring the continuity of the communication network throughout search operations,
  - enable the sharing of offline learning with other emergency services,

- strengthen the skills of drone operators and IT specialists, through accessible training programs, with a particular focus on artificial intelligence.

- **CT2MC** designs, produces, and operates USVs, robots, and applications to perform high-quality environmental sampling, monitoring, and inspection. With the participation to the chair, CT2MC expects:
  - To improve the interaction between a fleet of autonomous USVs and the human operator on the bank
  - To improve the efficiency of a fleet of USVs to perform collectively some task (water sampling, bathymetric measurement, pollutant monitoring, etc.)
  - To adapt the USV fleet organization to steer the connected UUV to its target while maintaining the wireless communication network connectivity
- **Thalès Land & Air Systems France** is interested to apply the approach and results of the chair related to emergent adaptive social organizations to anti-drone warfare (drone interception by a defense drone swarm), and to the deployment of enactive and physically informed mobile sensor networks.

IV. Education (2-3 pages)

The core of the educational programme associated with the FLOMAS chair concerns the teaching of AI at bachelor's degree level, more specifically within the framework of the bachelor's degree programmes in computer technology at the University Savoie Mont Blanc (Annecy University Institute of Technology) and the University Grenoble Alpes (Valence University Institute of Technology). The two institutes will share an educational engineer dedicated to AI, hired from January 2027 to December 2028 and funded by the chair, who will work part-time for the Annecy IUT (70%) and part-time for the Valence IUT (30%). Another important part of the chair's educational programme concerns continuing education and lifelong learning, mainly aimed at drone pilots and IT specialists from regional fire and rescue services. The third part of the chair's educational programme concerns students enrolled in the Master's programme in Applied Sciences at Grenoble INP – Esisar, located in Valence. The table below summarises the number of learners, learning hours and types of educational activities covered by the three parts of the educational programme.

	IUT Annecy (BSc)		IUT Valence (BSc)		SDIS 74 (LP)		G-INP Esisar (MSc)	
	hours	students	hours	students	hours	learners	hours	students
Creation of AI modules	230	55-70			30	15-30		
Adaptation of AI modules			30	40			84	30-40
AI projects	485	55-70	150	40			57	10

Number of learners, learning hours and types of educational activities

UNDERGRADUATE EDUCATION

University Bachelor of Technology In Computer Sciences (BUT Informatique)

The University Bachelor of Technology degree is structured around a nationally defined program. Institutions are granted a 30% margin for local adaptation, allowing for pedagogical innovation and alignment with emerging technological trends. We plan to revise the current program by integrating dedicated AI modules and projects. A new version of the French national BUT Informatique program is scheduled for implementation in September 2027. While the framework is still under development, several AI-related competencies have already been defined, laying the groundwork for future teaching and research initiatives.

- **Savoie Mont-Blanc University - Institute of Technology (IUT Annecy) - BUT Informatique**

Objectives: Provide 130 hours of AI modules to third-year students and 100 hours to second-year students. Focus the theme of projects (Learning and Assessment Situations - SAé) on the intelligent analysis of data from sensors and IoT.
Context: The University Bachelor of Technology In Computer Sciences (BUT Informatique) program at the Annecy University Institute of Technology offers three specialisation tracks. These tracks begin in semester 4 (170 specific hours) and conclude in semester 6 (615 specific hours in the third year). Among them, the Data Administration, Management, and Exploitation (AGED) track is designed to train mid-level professionals capable of administering databases and handling large-scale datasets. This track is mainly focused on paradigms and technologies related to Business Intelligence. At present, the national program only includes a single 24-hour module in Artificial Intelligence.
The new BUT Informatique program will allow specialization tracks to begin as early as semester 3. In line with the requirements of the Ministry of Higher Education and Research, the AGED track will be

replaced by an AI-oriented track named *Artificial Intelligence and Data Engineering*, whose program is currently being developed.
Adaptation of the third-year curriculum for the AGED track (2027–2028): this project proposes adapting the third-year computer science program of this track to focus on AI techniques.

We plan to create four modules (resources), totalling approximately **130 hours**:

- Building and adapting AI models: implementing simple AI models by adapting existing models
- Testing and validating AI solutions: evaluating AI model performance and identifying and mitigating algorithmic biases to ensure model fairness
- Deploying and monitoring AI solutions: putting AI models into production and monitoring their performance
- Mathematics for AI

The SAé in semesters 5 and 6 (**285 hours**) will focus on applying AI techniques applied to sensor networks and IoT. 25~30 students by year will be directly impacted by this transformation.

Development of second-year resources for the "Artificial Intelligence and Data Engineering" track (2028–2029): the program for this track is still under development, but we estimate that approximately **100 hours of resources and 200 hours of additional SAé** will need to be created starting in September 2028. Project-based activities (SAé) will be oriented toward the same themes: AI applied to sensor networks ans IoT. Each year, 30~40 students will be directly impacted by this transformation.

This work will be developed with the support of **one pedagogical engineer (shared with IUT Valence)** and by the researchers working in these fields (Khadija Arfaoui, Argheesh Bhanot, Mickaël Bettinelli, Vincent Couturier, Christophe Lin-Kwong-Chon, Nicolas Méger).

In parallel, the Savoie Mont Blanc University has submitted an AMI CMA France 2030 project focused on AI and robotics. This project should enable us to acquire a powerful computing infrastructure to support the implementation of these various modules.

- **Grenoble Alpes University - Institute of Technology (IUT Valence) - BUT Informatique**

Objectives: Give general knowledge about AI to third year students. Give them practical experience with distributed intelligence and multi-agent systems.

Pedagogical form: The implementation of the new nationwide-defined study program of the BUT in computer science will be an ideal opportunity to develop new teaching modules dedicated to the envisioned topics. The teaching in this program is generally organized around three formats: lectures, tutorials, and practical sessions. These three formats are complemented by cross-disciplinary projects known as Learning and Assessment Situations (SAÉ). A new teaching unit dedicated to AI will be introduced in the third year. The general knowledge lectures will be proposed by Jean-Paul Jamont who has taught this topic for 15 years at several universities, mainly at the M2 level. These lectures will be completed by ones on more specific topics such as multi-agent systems which are also taught by the project members at the M2 level at the ESISAR school of engineering and are aligned with the research topics this project. These lessons will be completed with practical sessions on these specific topics.

An SAÉ on the topic of distributed intelligence and multi-agent systems will be constructed with the support of **one pedagogical engineer** (shared with IUT Annecy). This cross-disciplinary project will involve networking, databases, and web technologies to give students practical knowledge of distributed intelligent systems. This SAÉ will go beyond micro-services architectures by introducing collective intelligence algorithms allowing systems components to coordinate their actions to build new functionalities. Distributed and embedded systems have been taught at IUT Valence in the past because of the proximity with the LCIS laboratory research themes and local companies demands. The last nation-wide study program removed almost all teaching on these topics. The new one will give us the opportunity to initiate a re-orientation of some teaching units towards these topics and prepare students to contribute to the local business ecosystem. The construction of the SAÉ will be accompanied with grant applications for small embedded systems to deploy student work on and give them practical experience with these systems. This SAÉ will involve **~40 third-year students during ~150h**. It will remain a significant part of the diploma during several years, the general theme being adaptable to many example applications.

CONTINUING EDUCATION / LIFELONG LEARNING

Objectives: offer introductory sessions on artificial intelligence, then on multi-agent systems, and finally on multi-agent reinforcement learning (MARL and CFRL) approaches, dedicated to drone pilots and IT engineers from fire and rescue services and from our industrial network (including ODAS, CT2MC and Thalès LAS).

The first session will be dedicated to autonomous drone and drone swarm problems and to drone pilots (35~40 persons) and IT specialists **(**15~20 persons). We plan then to extend the formation to other offer regional emergency services, some of which are currently in the process of forming drone pilot teams (typically 20 pilots and 10 IT specialists for each department in the regional area.

The participation of Thalès LAS France engineers, both as learners or teacher is under discussion. We also plan to offer the AI modules developed for undergraduate students as part of our professional and lifelong learning programs.

GRADUATE EDUCATION (MSc in Applied Sciences)

MSc in intelligent secure systems (Diplôme d'Ingénieur, Grenoble INP - Esisar)

At Grenoble INP-Esisar, AI is currently being integrated into the curriculum through existing and new courses and projects, at various levels, combining theoretical (CM) and practical (TP) instruction. The educational program of the FLOMAS chair will in particular focus on the last year of the master programs (M2), on two courses whose program will change in September 2026 (at last September 2027) because of the current pedagogical reform of the whole Engineering training:

- [M2, 10-15 students, CM (27 hours), TP (15 hours)] - Distributed AI and Multi-agent Systems (CS534), which will cover federated learning and in multi-agent organizations
- [M2, 20-25 students, CM (27 hours), TP (15 hours)] – Complex Networks (AC562), which will cover mobile sensor networks and control theoretic analysis of organizations

The development in the FLOMAS chair will also be the opportunity of proposals for innovation projects:

- [M2, 10-15 students, 57 hours project] - Innovation Project (PX505), multidisciplinary teams work on developing prototypes or demonstrators in robotics, AI, embedded systems and cybersecurity (among 17 projects per year, with typically at least 2-3 projects concerned with distribued approaches for CPSs)

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Le Bourget du Lac, le 02 décembre 2025

Support letter for the FLOMAS Chair Proposal

Dear Madam, Sir,

I am writing to express the official of CT2MC for the chair proposal “Federated learning and organization in mobile agent societies” (FLOMAS), presented as part of the MIAI Cluster AI 2025-2026 program.

CT2MC designs, produces, and operates USVs, robots, and applications to perform high-quality environmental sampling, monitoring, and inspections. To minimize its environmental impact, CT2MC advances manufacturing techniques and develops processes and products that are more environmentally friendly. CT2MC operates in several key areas:
- Manufacturing robots for pipe inspections, particularly in wastewater systems - Developing, prototyping, and manufacturing robots with composite material integration
- Producing multiple ranges of aquatic drones for environmental monitoring and analysis:
  - *SPYBOAT® range: aerial propulsion drones*
  - *INN'EAU range: water-thrust propulsion drones, floats, etc.*
  - *CAN'EAU range: specialized for confined space inspections (pipes, tanks, etc.)*

- Providing inspection services for surface water and confined spaces using these various products

We consider the topics addresses in the FLOMAS chair proposal, coordinated by Prof. Laurent Lefèvre (LCIS lab, UGA Valence) and Assoc. Prof. Mickaël Bettinelli (LISTIC lab, USMB Annecy) to be highly relevant and ambitious. The project addresses open questions related to adaptive organizations and federated learning in mobile agent societies which are, to the best of our knowledge, beyond the state of the art and of great interest to us.

Our collaboration in this project will concern the deployment of a fleet of autonomous drones (USVs) to carry out bathymetric surveys or water sampling in natural environments (lakes, rivers, etc.). These types of missions require autonomous reconfiguration of the fleet, whether to maintain communication through the ad hoc RF communication network (limited range), to avoid fixed or mobile obstacles, or to track environmental dynamics (movement of pollutants, currents, etc.).

Our participation in this project would include providing data (past missions) and use cases, drones (USV) for field experiments, and joint supervision of a post-doc researcher (two years) with a research engineer at CT2MC, within the framework of the ANR PRCE project ROS-EAU (under submission, coordinated by Prof. Eric Duviella, IMT Nord Europe).

We fully support this FLOMAS project which has the potential to significantly advance the state of the art in the design autonomous adaptive social organization and federated learning for cyberphysical agent societies and offers excellent valorisation opportunities in the field of USVs supervision and coordination.

Yours sincerely,

Olivier LE MEAUX

Président Fondateur

O. Le Meaux

ODAS SOLUTIONS

UNIVERSITE SAVOIR MONT BLANC
University Institute of Technology
LISTIC Laboratory

9 Rue de l'arc en ciel
BP 240
74942 ANNECY LE VIEUX

Annecy, le vendredi 5 décembre 2025

Monsieur,

Je, soussigné Arnaud STEPHAN, Président au sein de l'entreprise ODAS SOLUTIONS, confirme avoir pris connaissance du projet FLOMAS porté par Laurent Lefèvre et Mickaël Bettinelli dans le cadre de l'appel à chaire MIAI.

Notre entreprise ODAS SOLUTIONS exprime son intérêt pour ce projet et souhaite s'y associer en qualité de partenaire. À ce titre, il accompagnera le consortium en mettant à disposition, selon les besoins du projet et la disponibilité de ces ressources :

- Les données métiers nécessaires à la conception et à la validation des travaux
- L'accès à des drones pour les phases de test et de validation
- Son expertise métier pour l'orientation et l'évaluation des contributions.

Nous confirmons ainsi notre soutien au projet et notre volonté de contribuer à son bon déroulement.

Pour faire valoir ce que de droit,

ODAS SOLUTIONS
Arnaud STEPHAN
Président

Lu et approuvé

Madame, Monsieur,

Je, soussigné(e) Lieutenant DARNE Stéphane, responsable de la spécialité DRONE au sein du **Service Départemental d'Incendie et de Secours de la Haute-Savoie**, confirme avoir pris connaissance du projet **FLOMAS** porté par **Laurent LEFEVRE** et **Mickaël BETTINELLI** dans le cadre de l'appel à chaire MIAI.

Le SDIS 74 exprime son intérêt pour ce projet et souhaite s'y associer en qualité de partenaire. À ce titre, il accompagnera le consortium en mettant à disposition, selon les besoins du projet et la disponibilité de ces ressources :

- les données métiers nécessaires à la conception et à la validation des travaux
- l'accès à des drones pour les phases de test et de validation
- son expertise métier pour l'orientation et l'évaluation des contributions.

Nous confirmons ainsi notre soutien au projet et notre volonté de contribuer à son bon déroulement.

Pour faire valoir ce que de droit,

Lieutenant DARNE Stéphane
Responsable de la spécialité DRONE
Service Départemental d'Incendie et de Secours de la Haute-Savoie

Le chef de centre du CS Cruseilles

Lieutenant Stéphane DARNE